

Autologous profiling reveals inter-patient heterogeneity in Vδ2⁺γδTIL responses to glioblastoma driven by extracellular matrix-BTN3A axis

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ABSTRACT

Background The effectiveness of immunotherapies against glioblastoma (GB) remains limited. A major obstacle in advancing new strategies is the reliance on non-autologous systems, which do not accurately mimic the true extent of inter-patient heterogeneity in both immune responses and tumor susceptibility. This often leads to misleading conclusions about therapeutic efficacy and targetability.

Methods In this study, we addressed this critical gap by employing a fully autologous model. We phenotypically characterized primary αβ and γδT cells from the peripheral blood and tumors of 40 brain tumor patients, including 36 with confirmed GB, and expanded and functionally assessed the autologous anti-GB reactivity in a subset of patients.

Results Notably, only Vδ2⁺ and Vδ2[−] γδT cells, but not αβT cells, recognized autologous tumors. While Vδ2[−] γδT cells showed activity in a subset of patients, Vδ2⁺ γδTILs from all patients responded to autologous GB cells in the presence of pamidronate. In patients, a higher percentage of Vδ2⁺ γδTILs was associated with longer overall survival. However, the potency of Vδ2⁺ γδTILs varied markedly between individuals, highlighting substantial inter-patient heterogeneity in γδT cell-mediated tumor recognition. This variability was driven by differences in both immune cell-intrinsic features and tumor-intrinsic factors, including expression of BTN2A1 and especially BTN3A, the ligands of the Vδ2⁺ γδTCR. Functional assays revealed that anti-GB reactivity was further modulated by stimulatory and inhibitory co-receptors such as NKG2D, CD94, and TIGIT. Transcriptomic analysis linked Vδ2⁺ γδT cell reactivity to extracellular matrix (ECM) pathways and disrupting ECM components such as LAMA5 and TGFβ1 enhanced T cell responses. Knockout of ITGA3, a LAMA5 receptor, increased BTN2A1 and BTN3A expression on GB cells, improving immune recognition.

Conclusions This study demonstrates that inter-patient heterogeneity in Vδ2⁺ γδTIL responses to GB is driven by the extracellular matrix-BTN3A axis. Autologous systems effectively capture this heterogeneity, offering a reliable platform to identify determinants of both immune function and tumor vulnerability, insights that are essential for the rational design of γδTIL-based immunotherapies.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Vδ2⁺ γδT cells have demonstrated cytotoxic potential against glioblastoma (GB) in preclinical studies; however, clinical translation has been hindered by the widespread use of non-autologous models that fail to capture the true extent of inter-patient variability in immune repertoire potency and tumor susceptibility.

WHAT THIS STUDY ADDS

⇒ While we observed that increased frequencies of Vδ2⁺ γδTILs are associated with improved overall survival in GB, our study goes further by identifying which patients are most likely to benefit from Vδ2⁺ γδT cell-based therapies. Using a fully autologous platform, rarely applied in GB immunotherapy research, we uncovered pronounced inter-individual heterogeneity in Vδ2⁺ γδTIL activity. This approach enabled the discovery of the extracellular matrix-BTN3A axis as a critical regulator of γδT cell function. Moreover, we demonstrate that therapeutic efficacy is not solely determined by cell abundance, but hinges on both ligand accessibility and effector cell quality.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ By exposing key immune and tumor-intrinsic barriers to γδT cell efficacy in GB, this work provides a mechanistic blueprint for rational, personalized immunotherapy. It supports patient stratification based on γδTIL functionality and BTN3A-axis engagement and calls for the integration of autologous models in preclinical pipelines to avoid misleading efficacy signals and accelerate clinical translation.

INTRODUCTION

Isocitrate dehydrogenase wild-type glioblastoma (GB) is a high-grade, highly aggressive, and common adult brain tumor. Despite current multimodal treatment, including surgical resection followed by radiotherapy and temozolomide chemotherapy, prognosis

remains poor, with a median survival of approximately 15 months in adults.¹ The intracranial location and presence of the blood–brain barrier also limit the efficacy of many systemic therapies.^{2,3} Harnessing tumor-infiltrating immune cells has shown clinical benefit in several cancer types.⁴ However, GB is considered a ‘cold’ tumor, characterized by a profoundly immunosuppressive microenvironment that restricts T cell infiltration.^{5,6} In addition, GB frequently lacks well-defined tumor-specific antigens.⁷ These barriers to antigen-dependent targeting highlight the need for alternative immune approaches, such as those involving $\gamma\delta$ T cells, that are not constrained by classical HLA-restricted recognition.

$\gamma\delta$ T cells have emerged as compelling effectors in cancer immunotherapy, with infiltration of $\gamma\delta$ T cells correlating with favorable prognosis across several solid tumors.^{8–11} Representing 1%–10% of circulating T cells, $\gamma\delta$ T cells are defined by their δ -chain usage. V δ 2⁺ $\gamma\delta$ T cells, typically paired with V γ 9, dominate in peripheral blood, whereas V δ 2[−] subsets (including V δ 1, V δ 3, and V δ 5) are more common in mucosal and epithelial sites.^{10–12} $\gamma\delta$ T cells include innate and adaptive immunity, and despite being less abundant and well-characterized than $\alpha\beta$ T cells, they possess potent anti-tumor activity, especially against HLA-deficient tumors.¹³ Notably, $\gamma\delta$ T cell recognition is subset-specific. V δ 2⁺ $\gamma\delta$ T cells recognize a phosphoantigen-regulated complex of butyrophilins (BTN2A1 and BTN3A), and their activity can be pharmacologically enhanced by aminobisphosphonates.^{14–19} In contrast, the antigens for V δ 2[−] T cells remain less defined, though they are known to engage structurally diverse targets such as annexin A2, CD1d, and EPCR.^{20–22} $\gamma\delta$ T cell function is further modulated by co-receptors, activating (eg, NKG2C, NKG2D, NKp30/44/46) or inhibitory (eg, NKG2A, PD-1, TIGIT), that fine-tune their antitumor responses.^{12,23–26}

On recognition, $\gamma\delta$ T cells rapidly secrete inflammatory cytokines and execute cytotoxic responses against a broad range of tumors.²⁷ Despite this potential, the phenotypic and functional characteristics of $\gamma\delta$ tumor-infiltrating lymphocytes ($\gamma\delta$ TILs) in GB remain poorly defined. While prior studies have shown that V δ 2⁺ $\gamma\delta$ T cells are prevalent in the blood and brains of GB patients²⁸ and that expanded V δ 2⁺ $\gamma\delta$ T cells from healthy donors can target GB in vitro²⁹ and in vivo,³⁰ these findings rely primarily on non-autologous systems. Such approaches overlook the extensive inter-patient heterogeneity in both immune reactivity and tumor susceptibility, leading to potentially misleading conclusions. In particular, the relevance of these findings to actual patient tumors remains unclear without the use of autologous material. To address this critical bottleneck, we established a unique biobank of freshly resected GB tumors and matched autologous TILs. Our platform enabled in-depth characterization of distinct T cell subsets, with the aim of identifying strategies that account for patient-specific variability and improving the precision of $\gamma\delta$ T cell-based immunotherapies for GB.

METHODS

Patient-derived tumor material

Primary tumor pieces were mechanically fragmented into smaller pieces (~2 mm), followed by an enzymatic digestion using the human Tumor Dissociation Kit (Miltenyi) and a gentleMACS Octo Dissociator (Miltenyi) according to the manufacturer's instructions. Dissociated tumor suspension was either cultured, analyzed directly, or cryopreserved at −80°C in freezing medium containing 90% fetal calf serum (FCS; Sigma-Aldrich) and 10% dimethylsulfoxide (DMSO; Sigma-Aldrich).

For GB tumor cell cultures, tumor samples were fragmented into smaller pieces (~2 mm) and cultured in DMEM medium (Gibco) supplemented with 10% FCS, penicillin (100 IU/mL, ThermoFisher), and streptomycin (100 µg/mL, ThermoFisher) at 37°C, 5% CO₂. Medium was refreshed weekly and cells were passaged on reaching 80% confluence using 0.5% trypsin-EDTA (Gibco). Tumor samples were characterized by their morphology under a Brightfield microscope, and GB subtypes were classified morphologically and selected samples (P01, P02, P03, P05, P07) subsequently verified by bulk RNA sequencing for the expression of GB subtype-associated genes.³¹

Immunohistochemistry

Tumor tissues were formalin-fixed, paraffin-embedded (FFPE), and sectioned into 3 µm slices. Immunohistochemical staining for CD3 and $\gamma\delta$ TCR was performed using a Ventana Benchmark Ultra Immunostainer (Roche Diagnostics). CD3 staining was carried out using protocol CC1 with a 32 min incubation of the 2GV6 clone antibody (Roche, ready-to-use kit). $\gamma\delta$ TCR staining employed the same protocol with a 60 min incubation using the H41 clone antibody (Santa Cruz, 1:50 dilution). Slides were scanned and imaged with a Hamamatsu NanoZoomer XR (Hamamatsu Photonics).

PBMC isolation

Human peripheral blood mononuclear cells (PBMCs) were isolated from healthy donor buffy coats (Sanquin Blood bank, The Netherlands) and peripheral blood from GB patients. Isolation of PBMCs was performed using density gradient centrifugation with Ficoll-Paque (GE Healthcare). Cells were either directly cultured or cryopreserved at −80°C.

Cell lines

The U87-MG cell line was obtained from ATCC and was cultured in DMEM medium supplemented with 10% FCS and 1% penicillin/streptomycin. Adherent cells were detached using 0.5% trypsin-EDTA. Cell lines were passaged twice a week and cultured at 37°C with 5% CO₂.

T cell culture

Patient-derived and healthy donor T cells were cultured using a 2-week rapid expansion protocol as described previously.³² In short, T cells were cultured in RPMI-1640 medium supplemented with 10% human serum,

penicillin (100 U/mL), streptomycin (100 µg/mL), and β-mercaptoethanol (50 µM, Merck) at 37°C with 5% CO₂. Biweekly stimulation of the cells was performed with IL-2 (50 U/mL, R&D Systems), IL-15 (5 ng/mL, R&D Systems), and phytohemagglutinin-L (1 µg/mL, Sigma-Aldrich), as well as an irradiated feeder cell mixture consisting of PBMCs (1×10⁶ cells/mL, 3500 cGy), LCL-EBV transformed B cells (0.1×10⁶ cells/mL, 8000 cGy), and Daudi cells (0.1×10⁶ cells/mL, 8000 cGy). Medium was refreshed with the addition of IL-2 (50 U/mL) and IL-15 (5 ng/mL) every three to 4 days. Cells were rested without cytokines 3 days prior to experiments.

Flow cytometry

The antibodies that were used for flow cytometry are mentioned in online supplemental Method Table 1. Antibodies were diluted in PBS according to manufacturer's instructions, stained for 30 min at RT, washed with PBS, and fixed in 1% paraformaldehyde. Data were acquired using FACS Canto (BD Biosciences) or LRS Fortessa (BD Biosciences), and data were analyzed using FACS Diva software (BD Biosciences).

FACS sorting

Patient-derived T cells were FACS sorted into different T cell bulk populations using the Aria II (BD Biosciences). Cells were stained for CD3, CD8α, αβTCR, γδTCR, Vδ1, and Vδ2 (see Method table 1 in online supplemental Methods). After staining, samples were washed with PBS and resuspended in PBS containing 10% FCS and subsequently sorted into distinct groups: αβT, Vδ2⁺ γδT, Vδ1⁺ γδT, and Vδ1⁻ Vδ2⁻ γδT cell bulks (online supplemental figure 1A). After FACS sorting, cells were put on a rapid expansion protocol as described above, with one exception: on the day of sorting, an additional 2% of penicillin/streptomycin was added to avoid potential risks of infections.

IFNγ release ELISpot

Interferon γ (IFNγ) release ELISpot assays were performed to evaluate the in vitro tumor reactivity of patient-derived T cell bulks against primary GB tumor cells according to the manufacturer's instructions (Mabtech). Detailed description can be found in online supplemental Methods.

Cytotoxicity assay and correlation with IFNγ release ELISpot

Information on the cytotoxicity assay and correlation between cytotoxicity and IFNγ ELISpot can be found in online supplemental Methods.

CRISPR Cas9 knock-out (KO)

Information on CRISPR Cas9 KO can be found in online supplemental Methods.

RNA seq library preparation and sequencing

cDNA libraries were prepared from total RNA derived from different primary GB tumors preincubated with 30 µM PAM using reverse transcription and PCR amplification

(Single Cell Discoveries). Libraries were sequenced on an Illumina NovaSeq X Plus platform. Aligned and quantified gene-level counts were normalized using the DESeq2 package in R, applying size-factor normalization to adjust for library size differences.

Correlation analysis and pathway enrichment

For correlation analysis between gene expression and clinical or experimental metrics, the Pearson correlation coefficient was calculated using the *cor.test* function in R. Genes with correlation coefficients above 0.8 or below -0.8 were then filtered based on their variance, with only those exceeding the 30th percentile selected for further investigation. Pathway enrichment tool *Enrichr* was used to cluster the anti-correlated genes into overlapping pathways for further selection.³³ For this, the interactive pathway description tool, *Reactome 2022*, was consulted. From this dataset, pathway terms with a significant adjusted p value (p<0.05) were selected and (a selection of) the genes corresponding with these significant terms were chosen for later functional validation.

Reference single-cell RNA sequencing data analysis using the GB cell atlas

To validate the expression of target genes identified in our dataset, we used the publicly available GB cell atlas.³⁴ This atlas, which provides single-cell RNA sequencing data of GB tumors across various sources, was accessed from Cellxgene. To visualize gene expression levels across cell types, *Seurat* and *ggplot2* R packages were used.

Statistical analysis

Statistical analysis was performed using GraphPad Prism V.10 for Windows (GraphPad Software). Groups were compared for statistical significance using a t-test, one-way or two-way analysis of variance, or non-parametric tests, which is indicated in figure legends. Bars represent means and error bars represent SDs. P values less than 0.05 were considered statistically significant and are indicated as follows: *p<0.05; **p<0.01; ***p<0.001 and ****p<0.0001.

RESULTS

Composition of GB-infiltrating T cells

In order to examine the presence and role of different T cell subsets in GB, tumor tissue of 40 brain tumor patients, of which 36 were confirmed GB (table 1) was digested and phenotypically characterized using two flow cytometry panels (gating strategies in online supplemental figure 1A-C). We collected and analyzed PBMCs from 33 of the 40 patients, including 30 with confirmed GB. The vast majority of T cells derived from PBMCs and TILs were αβT cells (>95%, figure 1A). Immunohistochemistry analysis of GB tissue slides of a select number of

Table 1 Patient characteristic

Characteristic		Overall
Number of patients		40
Gender (%)	F	14 (35.0)
	M	26 (65.0)
Age (mean (SD))		62.50 (12.51)
Classification (%)	Astrocytoma WHO grade 4	1 (2.5)
	Glioblastoma WHO grade 4	36 (90.0)
	Oligodendroglioma WHO grade 2	1 (2.5)
	Oligodendroglioma WHO grade 3	2 (5.0)
MGMT methylation (%)	Hypermethylated	7 (17.5)
	Non-methylated	16 (40.0)
	Unknown	17 (42.5)
Volume preoperative (mm ³ , mean (SD))		41.76 (38.62)
Volume post op (mm ³ , mean (SD))		6.64 (18.80)
Extent of resection (%) (mean (SD))		90.34 (11.66)
KPS preoperative (mean (SD))		80.86 (11.47)
Tumor location by hemisphere (%)	Bifrontal	1 (2.5)
	Left	14 (35.0)
	Right	25 (62.5)
Type of surgery (%)	Asleep	24 (60.0)
	Awake	16 (40.0)
Survival (days) (mean (SD))		366.70 (214.90)

A total of 40 brain tumor patients were included in this research, of which 36 were confirmed GB patients.
GB, glioblastoma; KPS, Karnofsky Performance Scale; MGMT, O6-methylguanine-DNA methyl-transferase.

patients (n=5) confirmed the residency of both $\gamma\delta$ and $\alpha\beta$ T cells in the tumor tissue ([figure 1B](#)), with a significant predominance of $\alpha\beta$ T cells ([figure 1C](#)). Interestingly, for $\alpha\beta$ TILs, we observed an inverse ratio of CD4⁺ and CD8⁺ T cells (ratio 0.74) compared with patient-derived and healthy donor-derived PBMC counterparts (ratio ≥ 2.0) ([figure 1D](#)). This was accompanied by a decrease in the percentage of naïve CD4⁺ and CD8⁺ $\alpha\beta$ T cells in TILs, as well as an increase in the percentage of effector memory cells in both subsets ([figure 1E](#)). For $\gamma\delta$ T cells, a significantly decreased percentage was observed in TILs when comparing their frequency to healthy donor-derived PBMCs (2% vs 5%) ([figure 1F](#)). The relative contribution of each of the $\gamma\delta$ T cell subsets remained similar, with the predominant subset being V δ 2⁺ $\gamma\delta$ T cells for all investigated donors and compartments (~60%, [figure 1G](#)). Like $\alpha\beta$ TILs, a decrease in the percentage of naïve V δ 1⁺ and V δ 1⁻ δ 2⁻ $\gamma\delta$ TILs was observed, while V δ 2⁺ $\gamma\delta$ TILs closely resembled healthy donor and patient-derived PBMCs ([figure 1H](#)). In line with earlier reports, most V δ 2⁺ $\gamma\delta$ T cells had an effector-like phenotype^{35 36} in healthy donor and patient-derived PBMCs, which remained consistent in V δ 2⁺ $\gamma\delta$ TILs.

Immune coreceptor expression varies between different compartments

Next, we examined the presence and expression levels of defined inhibitory (NKG2A, PD-1) and activating (NKG2C, NKG2D, NKp30, NKp44, NKp46, DNAM-1, CD56) receptors on the T cell subsets. CD8⁺ $\alpha\beta$ T cells derived from TILs exhibited significantly increased levels of activating NK receptors NKG2D and NKp46 and exhaustion marker PD-1, compared with healthy donor CD8⁺ $\alpha\beta$ T cells, while no differences were seen for CD4⁺ $\alpha\beta$ TILs compared with PBMC-derived CD4⁺ $\alpha\beta$ T cells ([figure 2](#)). V δ 1⁻ δ 2⁻ $\gamma\delta$ TILs did not show any significant differences compared with healthy donor-derived V δ 1⁻ δ 2⁻ T cells, whereas V δ 1⁺ $\gamma\delta$ TILs were found to have higher levels of both activating receptor NKp46, as well as inhibitor receptor NKG2A. Prominent differences in cell surface expression were also found in the V δ 2⁺ subset, where NKG2A significantly decreased in V δ 2⁺ $\gamma\delta$ TILs when compared with healthy donor and patient PBMC-derived V δ 2⁺ $\gamma\delta$ T cells, while cytotoxicity marker DNAM-1 was significantly upregulated in V δ 2⁺ $\gamma\delta$ TILs ([figure 2](#)).

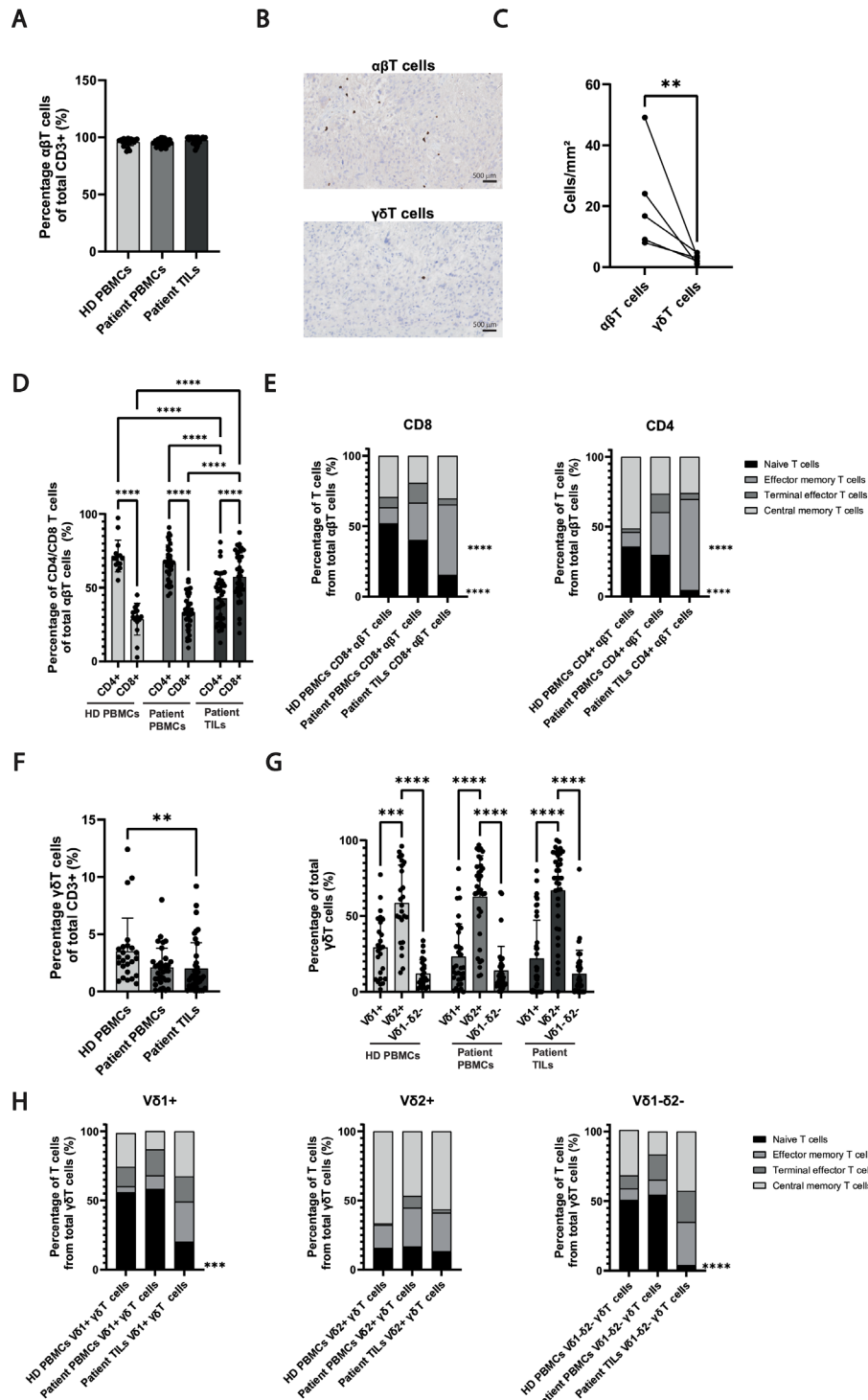


Figure 1 Phenotypic characterization of infiltrating T cell subsets in glioblastoma patients. (A) Percentage of $\alpha\beta$ T cells of total CD3+cells for healthy donor (HD) (n=25), patient-derived PBMCs (n=33), and TILs (n=40). (B) Immunohistochemistry images of infiltrating $\alpha\beta$ and $\gamma\delta$ T cells in GB tumor (n=5). (C) Quantification of infiltrating $\alpha\beta$ and $\gamma\delta$ T cells in GB (n=5). (D) Percentages of CD4⁺ and CD8⁺ of total $\alpha\beta$ T cells in HD (n=15), patient-derived PBMCs (n=33), and TILs (n=40). (E) CD4⁺ and CD8⁺ T cell subset expression of CD27 and CD45RA as markers for naive (CD27⁺, CD45RA⁺), terminal effector (CD27⁻, CD45RA⁺), effector memory (CD27⁻, CD45RA⁻) or central memory (CD27⁺, CD45RA⁻) phenotypes in HD (n=15), patient PBMCs (n=33) and TILs (n=40). (F) Percentage of $\gamma\delta$ T cells of total CD3+cells for HD (n=25), patient-derived PBMCs (n=33), and TILs (n=40). (G) Percentages of V δ 1⁺, V δ 2⁺ and V δ 1⁻2⁻ of total $\gamma\delta$ T cells in HD (n=15), patient-derived PBMCs (n=33), and TILs (n=40). (H) V δ 1⁺, V δ 2⁺ and V δ 1⁻2⁻ $\gamma\delta$ T cell subset expression of CD27 and CD45RA as markers for naive (CD27⁺, CD45RA⁺), terminal effector (CD27⁻, CD45RA⁺), effector memory (CD27⁻, CD45RA⁻) or central memory (CD27⁺, CD45RA⁻) phenotypes in HD (n=15), patient PBMCs (n=33) and TILs (n=40). Significance was calculated using a non-parametric Mann-Whitney U test (C) and two-way ANOVA (D–G) **p<0.01, ***p<0.001, ****p<0.0001. ANOVA, analysis of variance; GB, glioblastoma; PBMCs, peripheral blood mononuclear cells; TILs, tumor-infiltrating lymphocytes.

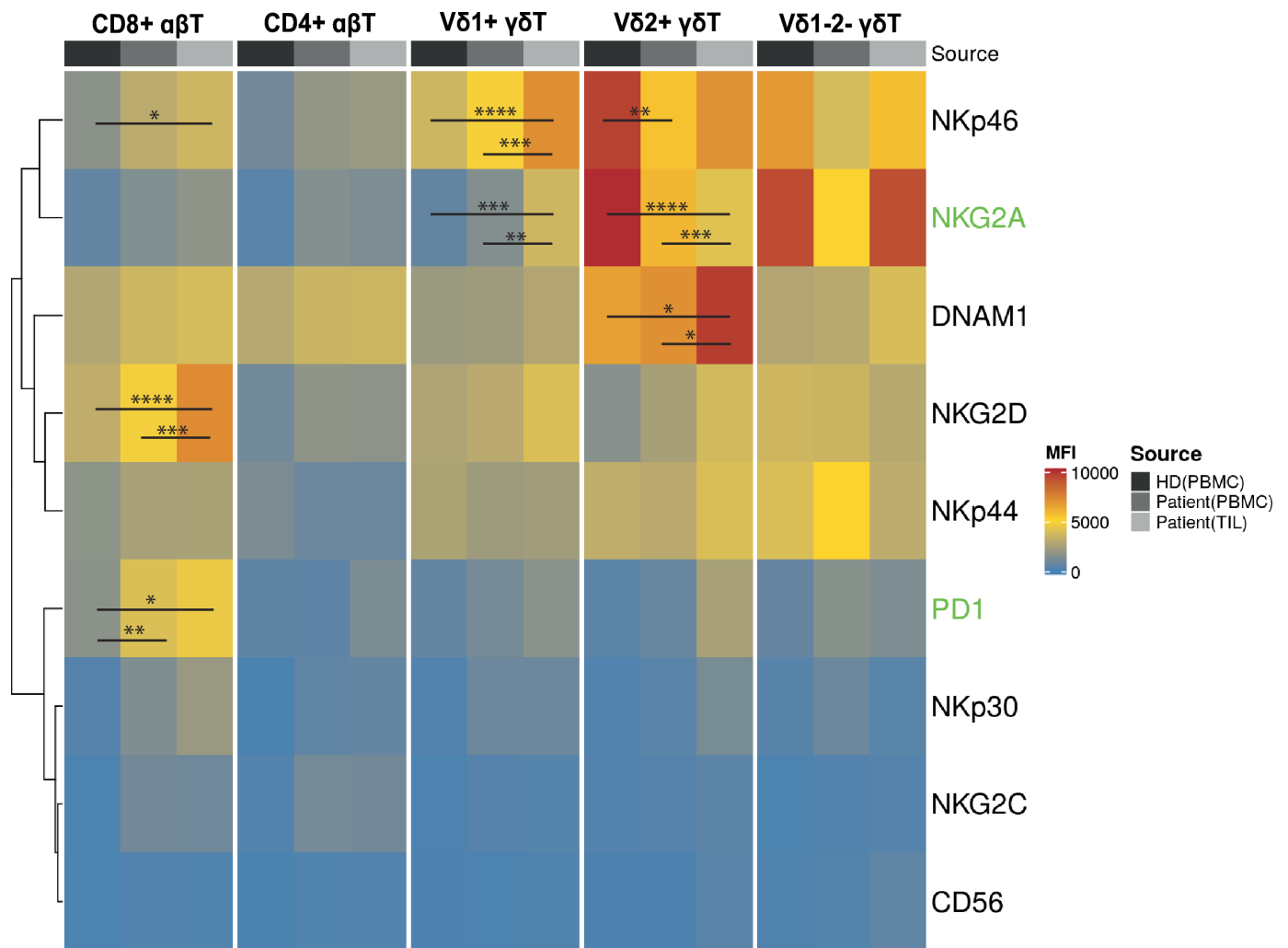


Figure 2 Cell surface protein levels of (innate) immune receptors across T cell subsets. Heatmap showing the Mean Fluorescent Intensity (MFI) values of distinct immune receptors across T cell subsets: CD8⁺ and CD4⁺ αβT cells, Vδ1⁺, Vδ2⁺ and Vδ1⁻2⁻ γδT cells, from different origins: healthy donor (HD) PBMCs (black, n=15), patient PBMCs (dark gray, n=33), and patient TILs (light gray, n=40). Receptors (rows) are indicated in green font for inhibitory types and in black font for activating types. Significance was calculated using a two-way ANOVA, followed by a post hoc test correcting for multiple comparisons, for each of the subsets and is indicated in the according conditions. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. ANOVA, analysis of variance; PBMCs, peripheral blood mononuclear cells; TILs, tumor-infiltrating lymphocytes.

Vδ1⁺ and Vδ1⁻δ2⁻ γδT cells derived from TILs show natural anti-GB reactivity against autologous tumors, but αβT cells do not

To determine whether and which fraction of the TILs are tumor-reactive and whether tumor-reactivity is increased in TILs compared with patient PBMC-derived T cells, we expanded patient-derived PBMCs and TILs in parallel to directly compare their phenotype and anti-GB reactivity using a rapid expansion protocol.³⁷ In addition, to benchmark the patient-derived T cell responses, we concurrently assessed the expansion and anti-GB reactivity of healthy donor PBMCs, as previously described by others.^{29,30} No significant differences were observed in absolute cell numbers when comparing expansion from donors or different compartments of patients until day 28. After day 28, αβT cells had a proliferation advantage when compared with γδT cells, resulting in higher numbers of αβT cells over time (online supplemental figure 2A-D).

Next, we tested the activity of expanded PBMC and TIL-derived T cell subsets from 8 patients (online supplemental table 1) against autologous tumors using an IFNγ ELISpot (figure 3A). Strikingly, no activity was observed from αβT cells, the most frequent subset, neither when they were isolated from PBMCs, nor when isolated from TILs (figure 3B), despite the high expression of MHC-I in most primary GB tumors (online supplemental figure 2E). In contrast, in 3 out of 6 tested patients, Vδ1⁺ γδTILs were active, arbitrarily defined as more than 30 spots per well, against autologous GB (figure 3B). For patient 06 and 07, we could test matched Vδ1⁺ γδT cells derived from PBMCs, which showed no activity. In one patient, P08, Vδ1⁻δ2⁻ γδT cells isolated from TILs showed activity against autologous tumors; however, no matched PBMCs were collected from this patient to assess whether this could also be seen in PBMC-derived T cells.

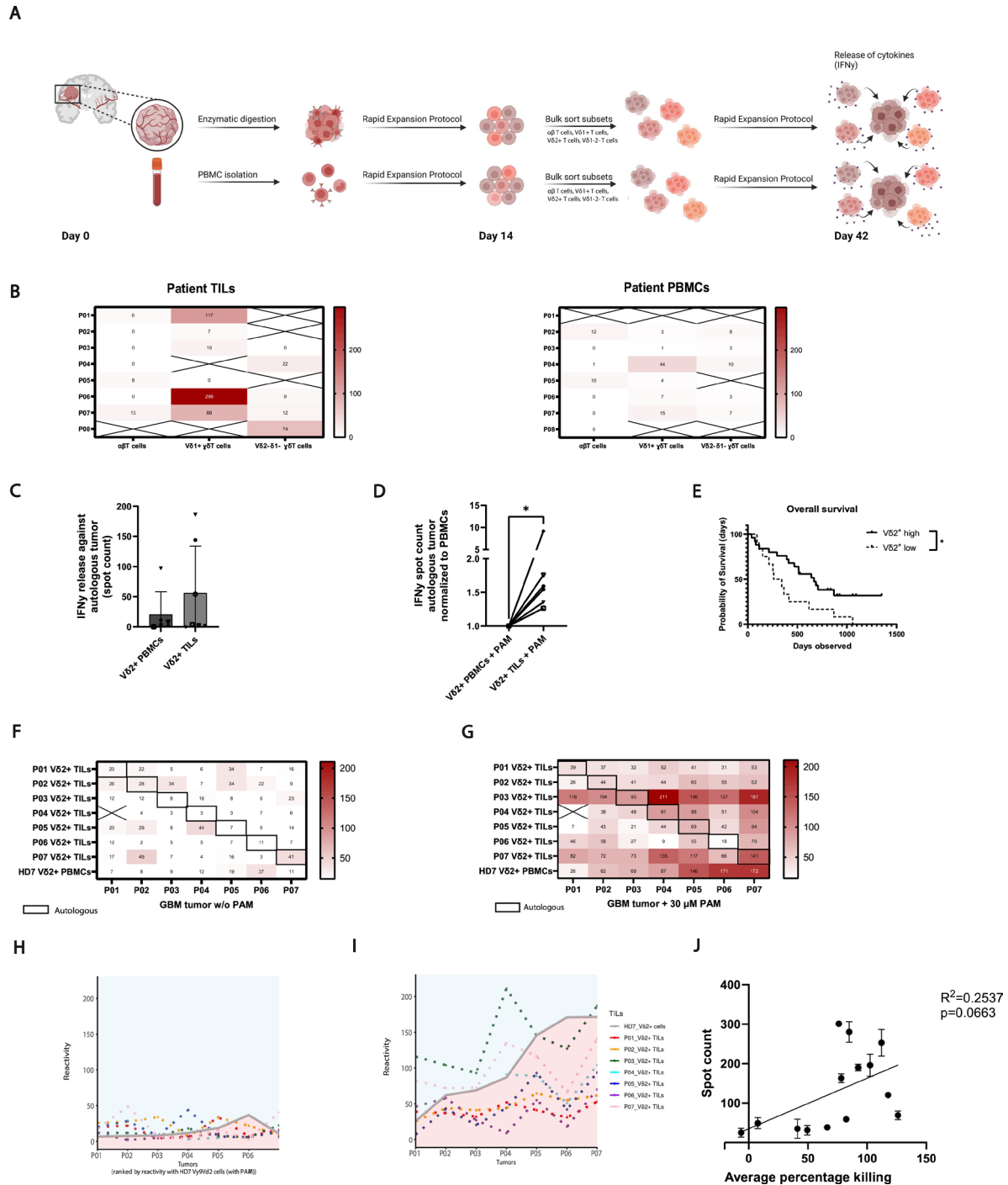


Figure 3 In vitro tumor-reactivity of T cell subsets against autologous and allogeneic GB tumors. (A) Experimental setup to evaluate in vitro expansion and GB-reactivity of various T cell populations. (B) IFN γ ELISpot count after overnight (20 hours) co-culture of $\alpha\beta$, V δ 1 $^{+}$ and V δ 1 $^{2-}$ $\gamma\delta$ T cells from match patient-derived PBMCs and TILs (n=8, all GB) and autologous tumor cells (E:T 3:1). (C) IFN γ ELISpot count after overnight (20 hours) co-culture of V δ 2 $^{+}$ $\gamma\delta$ T cells from match patient-derived PBMCs and $\gamma\delta$ TILs (n=6, all GB) and autologous tumor cells (E:T 3:1) with 30 μ M PAM. (D) IFN γ ELISpot ratio PBMCs:TILs after overnight (20 hours) co-culture of V δ 2 $^{+}$ $\gamma\delta$ T cells from match patient-derived PBMCs and $\gamma\delta$ TILs (n=6, all GB) and autologous tumor cells (E:T 3:1) with 30 μ M PAM. Dots represent different patients. (E) Probability of survival of total cohort (n=40) based on V δ 2 $^{+}$ percentages of total gdT cells. V δ 2 $^{+}$ high means >50% of $\gamma\delta$ T cells is V δ 2 $^{+}$ subset, V δ 2 $^{+}$ low means <50% of $\gamma\delta$ T cells is V δ 2 $^{+}$ subset. Survival was analyzed using a log-rank test. (F, G) IFN γ ELISpot count after overnight (20 hours) co-culture of V δ 2 $^{+}$ $\gamma\delta$ TILs and autologous and allogeneic GB tumor cells (E:T 3:1) in the absence (F) and presence of 30 μ M PAM (G) (n=2 independent experiments). (H, I) Different visualization of reactivity after overnight (20 hours) co-culture of V δ 2 $^{+}$ $\gamma\delta$ TILs and autologous and allogeneic GB tumor cells (E:T 3:1) in the absence (G) and presence of 30 μ M PAM (H). (J) Correlation of IFN γ release and cytotoxicity measured in coculture of autologous tumor cells with matched V δ 2 $^{+}$ $\gamma\delta$ TILs and HD V δ 2 $^{+}$ $\gamma\delta$ T cells (n=7 in two independent experiments). *p<0.05. GB, glioblastoma; PAM, pamidronate; PBMC, peripheral blood mononuclear cells; TIL, tumor infiltrating lymphocyte.

Anti-GB reactivity of Vδ2⁺ γδT cells and GB susceptibility to Vδ2⁺ γδT cells is heterogeneous between patients

Next, we explored the GB reactivity of matched PBMC- and TIL-derived Vδ2⁺ γδT cells from 6 patients, as the major subpopulation found in γδTILs, against matched primary GBs in the absence or presence of the amino-bisphosphonate pamidronate (PAM). In the absence of PAM, we observed a low reactivity of Vδ2⁺ γδT cells to GB and no significant difference in response by PBMC- and TIL-derived Vδ2⁺ γδT cells (figure 3C). However, in the condition with PAM, the relative IFNγ release of Vδ2⁺ γδTILs compared with their matched PBMC-derived Vδ2⁺ γδT cells was significantly enhanced for all six patients, though the increase in the strength of activity differed between patients (figure 3D, (online supplemental figure 2F)). Despite inter-patient variability in functionality, the dominance of Vδ2⁺ γδTILs, defined as Vδ2⁺ cells comprising more than 50% of the γδTIL population, was already associated with improved overall survival in GB patients from our cohort (figure 3E), underscoring the *in vivo* therapeutic potential of Vδ2⁺ γδTILs against GB tumors.

We then investigated in more detail the interindividual variability of Vδ2⁺ γδTILs against primary GB targets and focused on Vδ2⁺ γδTILs from 7 patients and Vδ2⁺ PBMCs from one healthy donor. In the absence of PAM, low activity of Vδ2⁺ γδTILs was observed against autologous GB (figure 3F) for only one patient (P07, Patients have been renumbered in this study based on recognizability (the higher the recognition, the higher the number) by healthy donor-derived Vδ2⁺ γδT cells). Interestingly, low activity against selected allogeneic GB could also be observed even when Vδ2⁺ γδTILs were not active against autologous tumor material (P01, P02, P05). Activity of PBMC-derived Vδ2⁺ γδT cells from a healthy donor was only observed against one primary GB and was low, in the absence of PAM. In contrast, in the presence of PAM, Vδ2⁺ γδTILs derived from 6 out of 7 patients were active against the autologous targets (figure 3G).

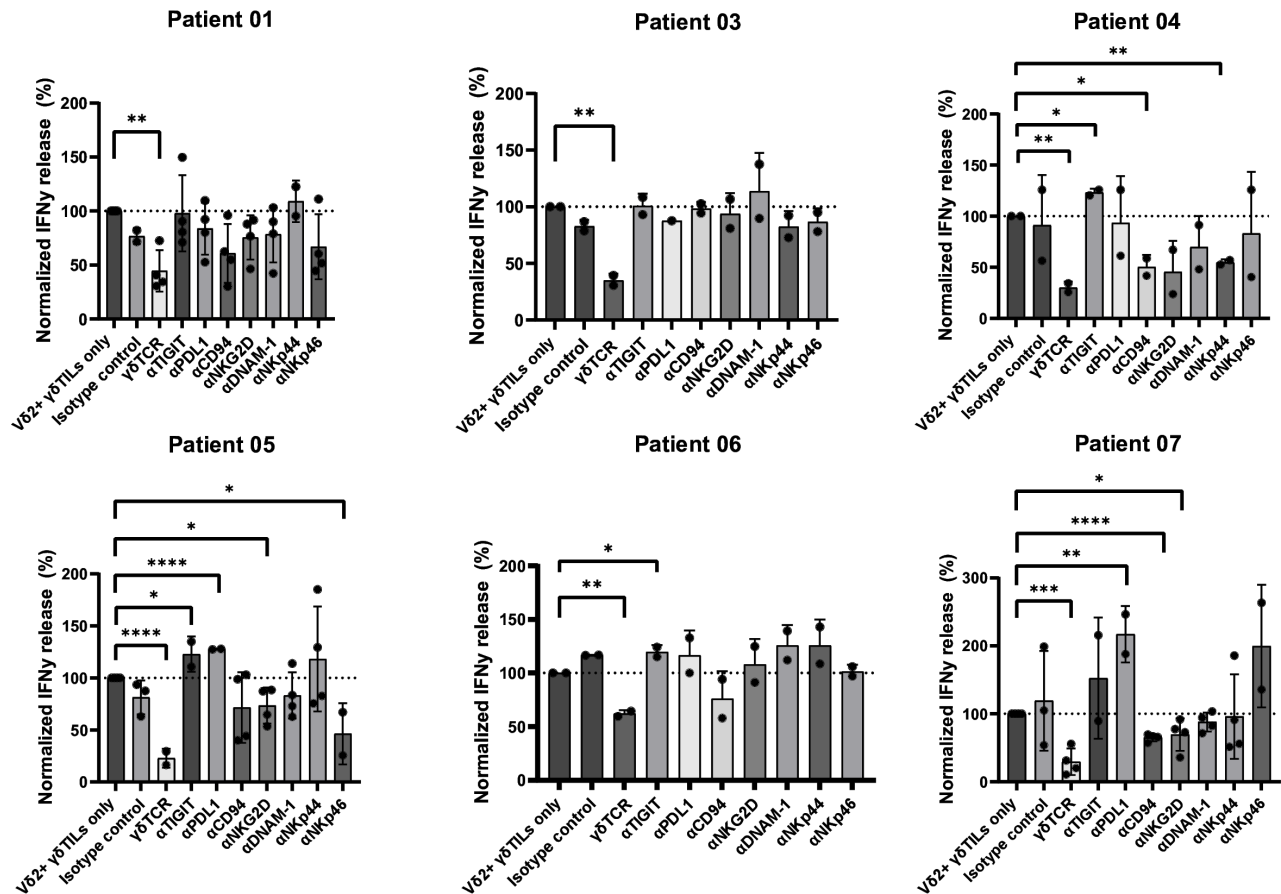
Heterogeneous reactivity against tumors was not only observed for autologous GBs as targets (range 18–172 spots) but also against allogeneic GB (range 7–211 spots). PBMC-derived Vδ2⁺ γδT cells from a healthy donor showed heterogeneous activity against different targets (range 26–172) in the presence of PAM as well, with the strongest activity against primary tumors derived from P05, P06, and P07. To further illustrate diversity in recognition, we ranked patients based on recognition assessed on cytokine secretion by healthy donor-derived Vδ2⁺ γδT cells (figure 1H,I). To formally assess whether ranking based on cytokine release more accurately reflects full T cell activation than killing, we performed IFNγ ELISpot and cytotoxicity assays in parallel. The experiment was conducted twice using four different primary GB tumors (P01, P03, P04, P07) with their corresponding autologous γδTILs, as well as the healthy donor HD7 Vδ2⁺ γδT cells. We confirmed effective killing of primary GB tumors and observed heterogeneous anti-GB reactivity among

autologous Vδ2⁺ γδTILs and healthy donor Vδ2⁺ γδT cells. Correlation of IFNγ release and cytotoxicity revealed that most Vδ2⁺ γδT cells exhibited concordant activity, either low or high in both assays, whereas a subset of Vδ2⁺ γδTILs or healthy donor Vδ2⁺ γδT cells efficiently killed GB cells without detectable IFNγ release (figure 3J). These findings indicate that cytokine release represents a more stringent marker of complete Vδ2⁺ γδT-cell activation than cytotoxicity alone. These data also illustrate substantial heterogeneity in the activity of Vδ2⁺ γδT cells against autologous and allogeneic primary GB tumors, likely driven by intrinsic differences in Vδ2⁺ γδT cells across individuals and by variation in ligand expression on GB cells, rather than by differences in GB tumor subtypes. The latter were assessed in selected patients (P01, P02, P03, P05, and P07) by bulk RNA sequencing, confirming also that the investigated samples represented bona fide GB encompassing classical, mesenchymal, neuronal, and proneuronal subtypes³¹ (online supplemental figure 2G).

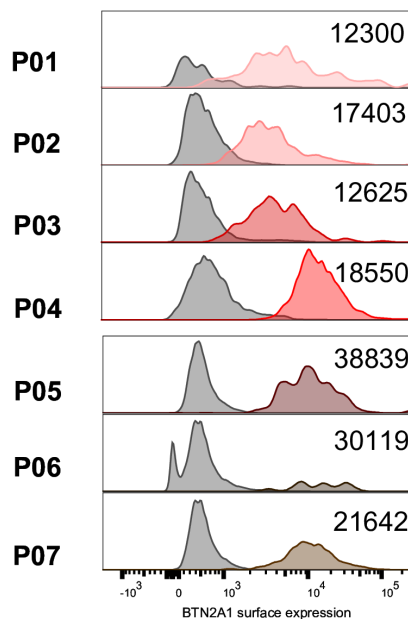
Anti-GB-reactivity of Vδ2⁺ γδTILs relies on their γδTCR and additional receptors that vary between patients

To dissect underlying molecular mechanisms of functional diversity of Vδ2⁺ γδT cells derived from different patients, we assessed activity of 6 GB-derived Vδ2⁺ γδTILs against GB in the presence of PAM, and in the presence or absence of blocking antibodies inhibiting specific (innate) receptors. To avoid a bias through diversity of the target such as differences in BTN2A1 and BTN3A expression, or other activating or inhibiting receptors, we used the GB cell line U87-MG as the target and measured the release of IFNγ on co-culture. Blocking the γδTCR resulted in a significant reduction of IFNγ production by Vδ2⁺ γδTILs of all tested patients. While for 2 patients the γδTCR was the only receptor that significantly dictated recognition of U87-MG by Vδ2⁺ γδTILs (P01 and P03), all other Vδ2⁺ γδTILs could be modulated by multiple receptors (figure 4A). For instance, blocking the inhibitory receptor TIGIT resulted in a significant increase of IFNγ levels produced by Vδ2⁺ γδTILs of patients P04, P05, and P06. Blocking PD-L1 showed significant enhanced activity for Vδ2⁺ γδTILs of patients P05 and P07 (figure 4A). Inhibition of CD94 on Vδ2⁺ γδTILs, which is able to dimerize with both the activating receptor NKG2C, as well as the inhibitory receptor NKG2A, showed a significant reduction in IFNγ release for patients P04 and P07, suggesting that the activation of Vδ2⁺ γδTILs of these patients could partly be modulated by the activating receptor NKG2C (figure 4A); however, surface expression of CD94 ligand HLA-E was rather low on all GB tumors (online supplemental figure 2H). Blocking of the activating receptors NKG2D, Nkp46, and/or Nkp44 decreased recognition in Vδ2⁺ γδTILs of patients P04, P05, and P07.³⁸ These data imply that in most patients, GB-derived Vδ2⁺ γδTILs use, in addition to the main Vδ2⁺ γδTCR driven activation, different mechanisms to modulate activity against GB in the presence of PAM (figure 4A).

A



B



C

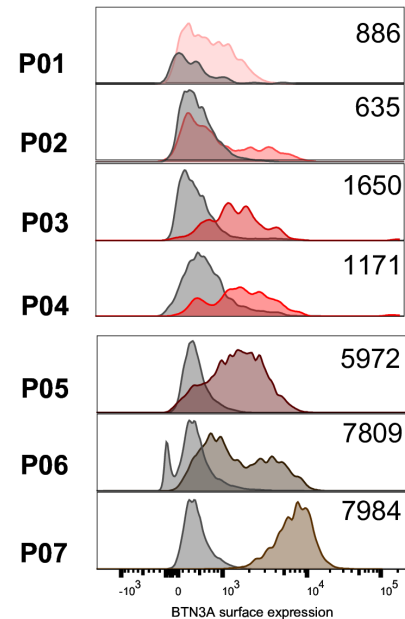


Figure 4 Deciphering potential mechanisms of V δ 2⁺ γ δ T cell–GB tumor interactions. (A) IFN γ ELISpot count normalized to only γ δ TILs condition after overnight (20 hours) co-culture of V δ 2⁺ γ δ TILs and U87-MG cell line (E:T 1:3, n=6 patients, with 2–4 replicates per patient) in the presence of 10 μ M PAM. T cells were preincubated with different blocking antibodies (10 μ g/mL final concentration for all antibodies) for 1 hour prior to co-culture. An unpaired t-test was used to test significance against the only γ δ TILs condition. (B) MFI numbers and flow cytometry plots of cell surface expression of BTN2A1 on GB tumor samples (n=5). (C) MFI of cell surface expression of BTN3A was assessed using flow cytometry on GB tumor samples (n=5). *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. GB, glioblastoma; MFI, mean fluorescent intensity; PAM, pamidronate.

Low BTN2A1 and BTN3A expression associates with poor tumor-reactivity of Vδ2⁺ γδTILs

To assess whether the observed variation in reactivity of healthy donor-derived Vδ2⁺ γδT cells and Vδ2⁺ γδTILs was mediated by differences in expression of the Vδ2⁺ γδTCR ligands at the cell membrane of against autologous GB, we measured expression levels of BTN3A and BTN2A1 on GB from seven patients, using flow cytometry. Tumors that were less recognized by Vδ2⁺ γδT cells (P01, P02, P03, and P04) showed a reduced and more heterogeneous expression of BTN2A1 when compared with well-recognized tumors (P05, P06, and P07, [figure 4B](#)). BTN3A surface expression was low, or partially negative in poorly recognized tumors, and was high in well-recognized tumors ([figure 4C](#)). Addition of PAM did not enhance expression levels of either BTN2A1 or BTN3A in selected patients (online supplemental figure 2I), implying that PAM merely drives the phosphoantigen-mediated complexation of BTN2A1 and BTN3A. In conclusion, BTN2A1 can be found at the cell membrane of all investigated GB samples, though at varying levels. BTN3A expression was more compromised in less recognized GB tumors, implying that BTN3A might be more limiting for recognition by Vδ2⁺ γδT cells than BTN2A1.

Extracellular matrix-related genes are involved in Vδ2⁺ γδT cell-mediated recognition of GB tumors

Since the recognition of GB tumors has been influenced not only by the Vδ2⁺ γδTCR and variations in their (innate) receptor expression, we hypothesized that additional molecules also contribute to the strength of immune responses against primary GB. To further explore this, we reanalyzed bulk RNA-seq data from five GB samples (P01, P02, P03, P05, P07), previously used for cell identity verification. Our aim was to identify genes in these GBs potentially associated with recognition (and activity) by Vδ2⁺ γδT cells. For this, we ranked the tumors based on their recognizability by Vδ2⁺ γδT cells ([figure 3F H](#)), specifically using the activity scores of healthy donor-derived Vδ2⁺ γδT cells to minimize variability in T cell reactivity. In order to identify genes that are significantly associated with a specific reactivity profile, we performed a combined correlation (online supplemental figure 3A) and variance filtering approach. Here, we identified a list of genes that were either highly positively correlated (266 genes) or negatively correlated (235 genes) with the reactivity profile ([figure 5A](#), online supplemental table 1 for the full gene list). From a therapeutic perspective, we focused on genes that were upregulated in less-recognized tumors (anti-correlated genes), with the aim of targeting these genes to improve tumor recognition. Pathway enrichment analysis identified terms related to extracellular matrix (ECM) organization and TGFβ signaling, revealing 17 connected genes ([figure 5B](#)). We also identified genes that are involved in semaphorin interactions, which have been shown to be involved in neuronal development, axon targeting, and synapse formation.^{39 40} As these genes lack a direct interaction with the genes from

the other pathways, we decided to focus on ECM and TGFβ-associated genes.

To validate the role of the ECM and TGFβ pathways, a selection of 6 genes (LAMA5, TGFB1, ITGA3, COLGALT1, SMURF1, ITGA11), that were identified using the pathway enrichment tools, were tested in vitro. We first verified the expression of the connected genes in GB tissue by referencing a GB scRNA-seq atlas, confirming that these genes were expressed by the tumor cell population ([figure 5C](#), online supplemental figure 3B). Although TGFB1 expression in the GB atlas was also observed in immune compartments, its detection in malignant clusters together with our bulk RNA-seq data suggested a potential tumor-intrinsic contribution. When examining the expression of these genes per patient in the GB atlas, we observed considerable interindividual heterogeneity ([figure 5D](#)). Next, we co-cultured healthy donor-derived Vδ2⁺ γδT cells with primary GB derived from a patient that showed low reactivity (P02) in the absence or presence of an α-LAMA5 or α-TGFβ1 antibody, to verify whether blocking of these genes could improve recognition. Indeed, inhibition of LAMA5 and TGFB1 shows an enhanced reactivity by Vδ2⁺ γδT cells against P02 primary GB cells ([figure 6A](#)). To validate this finding, these two genes were knocked out in a GB cell line U87-MG using CRISPR Cas9 (knock-out efficiency in online supplemental figure 4A). When the KO cells were co-cultured with Vδ2⁺ γδT cells derived from a healthy donor and patient TILs, we noticed a significant two-fold increase in IFNγ spots ([figure 6B](#)). This indicates the importance of the investigated genes for the recognition of GB cells by Vδ2⁺ γδT cells and highlights a potential therapeutic opportunity to increase the recognizability of GB that is difficult to target. Together, these data support a tumor-intrinsic, immunosuppressive role for TGFB1 and LAMA5 relevant to Vδ2⁺ γδT cell activity.

ITGA3 modulates tumor escape via downregulation of BTN2A1 and BTN3A

When individually knocking out the remaining four genes from our initial selection (ITGA3, COLGALT1, ITGA11, SMURF1), and the Vδ2⁺ γδTCR ligand BTN2A1 in U87-MG cells (online supplemental figure 4A), we observed a significant increase in the reactivity of healthy donor-derived Vδ2⁺ γδT cells, only in the case of integrin-α3 (ITGA3) KO cells. The KO of BTN2A1, as a positive control, resulted in the loss of Vδ2⁺ γδT cell activation ([figure 6C](#)). The use of Vδ2⁺ γδTILs from P07 as effector cells against the U87-MG ITGA3 KO cells confirmed that increased recognition does not depend on the donor or compartment from which Vδ2⁺ γδT cells were isolated ([figure 6D](#)). We verified these findings using patient-derived ITGA3 knockout (KO) cells from a case with previously poor recognition (P02). Following ITGA3 KO, P02 tumor cells showed a clear trend toward improved recognition ([figure 6E](#)). Furthermore, we investigated whether any of the functionally validated genes LAMA5, TGFB1, ITGA3 could have any prognostic value for GB

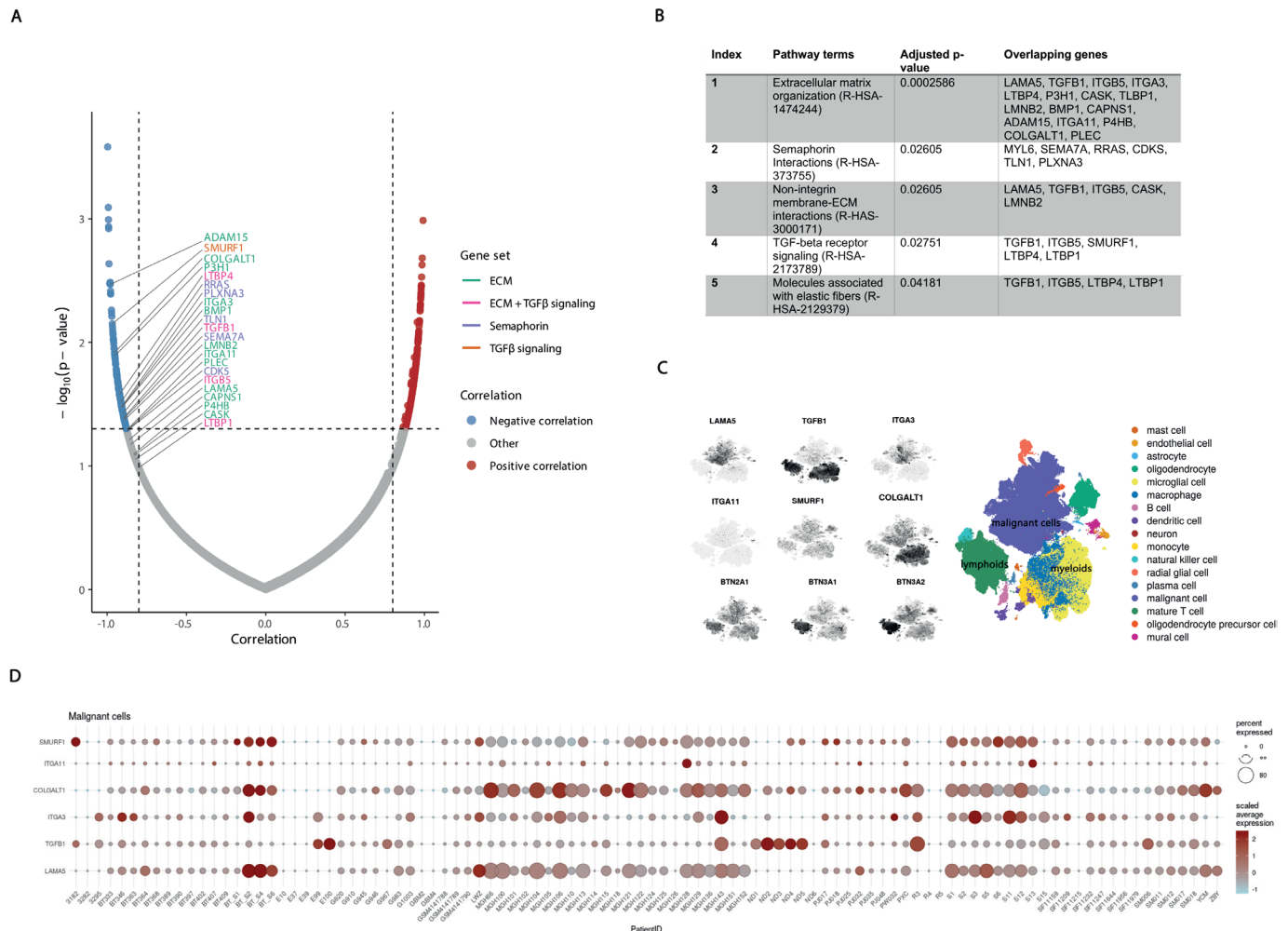


Figure 5 Identifying key genes in $V\delta 2^+$ $\gamma\delta T$ cell-mediated reactivity against GB tumors. (A) A volcano plot showing correlation coefficients (x-axis) and $-\log_{10}(p\text{-values})$ (y-axis) for gene expression levels correlated with healthy donor $V\delta 2^+$ $\gamma\delta T$ cell-mediated tumor recognition. Each point represents a gene. Selected genes from predefined functional groups (ECM, TGF β , and Semaphorin pathways) are highlighted and labeled. Genes are colored according to their functional grouping, and their position relative to the correlation threshold (± 0.8) and significance threshold ($p < 0.05$) is shown. (B) Pathway enrichment analysis of the anti-correlated gene set, showing significantly associated Gene Ontology (GO) terms and the corresponding genes within these pathways. (C) Expression levels of selected genes within a published GB cell atlas along with cell type annotation. (D) Expression of selected genes across individual patients within the ‘malignant cells’ subset of the GB cell atlas, highlighting patient-specific heterogeneity. GB, glioblastoma; ECM, extracellular matrix.

patients. Using the gene expression profiling interactive analysis (GEPIA2) dataset which divides 162 GB patients from The Cancer Genome Atlas (TCGA) into a high and low expression group ($n=81$ each), a log-rank test revealed that high ITGA3 expression correlated with significantly lower overall survival ($p=0.021$) (online supplemental figure 4B), while no survival differences were observed for LAMA5 or TGFβ1 (online supplemental figure 4C and D). We then wondered whether the expression of extracellular matrix genes LAMA5, TGFβ1, and ITGA3 could be correlated to the various $V\delta 2^+$ $\gamma\delta T$ cell mechanisms found in figure 4A. Indeed, we observed a significant correlation of LAMA5 and ITGA3 with $\gamma\delta TCR$ blocking and a nearly significant correlation of TGFβ1 and $\gamma\delta TCR$ blocking, but not with any of the other receptors (online supplemental figure 4E-G).

To reveal whether knocking out functionally confirmed genes also interferes directly with expression of $V\delta 2^+$ $\gamma\delta TCR$ ligands, we assessed the surface expression of BTN2A1 and BTN3A on U87-MG KO cells of our validated candidates. Interestingly, we witnessed elevated levels of both BTN2A1 and BTN3A on the ITGA3 KO, which could not be attributed to the KO procedure as different KO cell lines, COLGALT1, TGFβ1, and LAMA5, did not show this increase (figure 6E,F), suggesting a direct connection between BTN2A1 and BTN3A expression and ITGA3 and a potential manner of escaping $V\delta 2^+$ -mediated GB recognition for high-ITGA3 expressing GB.

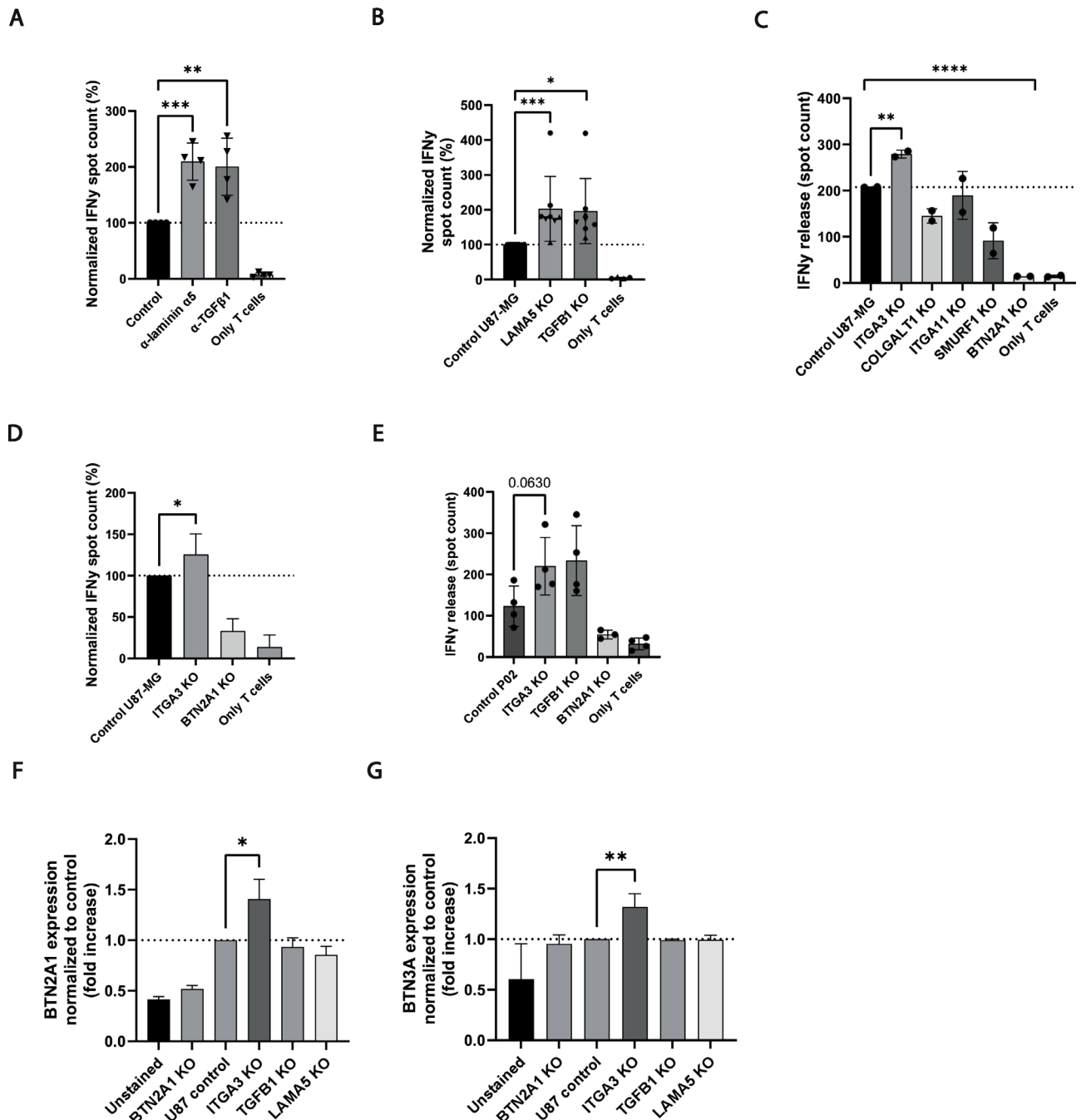


Figure 6 In vitro validation of selected ECM-related genes. (A) IFN γ ELISpot count normalized to only TILs condition after overnight (20 hours) co-culture of healthy donor V $\delta 2^+$ $\gamma\delta$ T cells and P02-derived GB tumor cells (E:T 3:1, n=2 independent experiments with two replicates per experiment) in the presence of 10 μ M PAM. (B) IFN γ ELISpot count normalized to U87-MG control condition after overnight (20 hours) co-culture of healthy donor V $\delta 2^+$ $\gamma\delta$ T cells (triangle) and P07 V $\delta 2^+$ $\gamma\delta$ TILs (circle) with different U87-MG KO cells (E:T 1:3, n=2 independent experiments with two replicates per experiment) in the presence of 10 μ M PAM. (C) IFN γ ELISpot count normalized to U87-MG control condition after overnight (20 hours) co-culture of healthy donor V $\delta 2^+$ $\gamma\delta$ T cells with different U87-MG KO cells (E:T 1:3, n=2 independent experiments) in the presence of 10 μ M PAM. (D) IFN γ ELISpot count normalized to U87-MG control condition after overnight (20 hours) co-culture of P07 V $\delta 2^+$ $\gamma\delta$ TILs with different U87-MG KO cells (E:T 1:3, n=3 with two replicates per patient) in the presence of 10 μ M PAM. An unpaired t-test was used to test significance in the functional validation assays against the U87-MG and P02 tumor control conditions. (E) IFN γ ELISpot count normalized to P02 tumor control condition after overnight (20 hours) co-culture of P07 V $\delta 2^+$ $\gamma\delta$ TILs and healthy donor V $\delta 2^+$ $\gamma\delta$ T cells with different P02 KO cells (E:T 1:3, n=2 with two replicates per patient) in the presence of 10 μ M PAM. An unpaired t-test was used to test significance in the functional validation assays against the U87-MG and P02 tumor control conditions. (F) MFI of cell surface expression of BTN2A1 was assessed using flow cytometry on several U87-MG KO cells (n=3 experiments). (G) MFI of cell surface expression of BTN3A was assessed using flow cytometry on several U87-MG KO cells (n=3 experiments). *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. ECM, extracellular matrix; GB, glioblastoma; MFI, mean fluorescent intensity; PAM, pamidronate; TILs, tumor-infiltrating lymphocytes.

DISCUSSION

Translating cancer immunotherapies to tumors like GB has proven challenging. Often due to a lack of predictive models that capture the complex and heterogeneous interactions between immune cells and their autologous tumor targets. Without autologous systems, critical variability is obscured, leading to flawed assumptions about immune competence and tumor susceptibility. To overcome this bottleneck, we established a comprehensive biobank of GB tumors and matched immune cells from 36 patients, enabling detailed phenotypic and functional analysis of $\gamma\delta$ TILs in a fully autologous setting. Our findings show that $\gamma\delta$ T cells were the only T cell population within the examined TILs capable of targeting autologous GB, and we identified multiple patient-specific and tumor-intrinsic factors influencing their activity. These insights lay the groundwork for rational clinical trial design that reflects the true biological complexity of GB.

Unlike most solid tumors,^{10,12} GB was predominantly infiltrated by $V\delta 2^+$ rather than $V\delta 2^-$ $\gamma\delta$ T cells, likely reflecting the tumor's strong vascular component. To date, $V\delta 2^-$ $\gamma\delta$ T cells have not been functionally studied in GB due to their low frequency.⁴¹ Nonetheless, within our autologous cohort, we identified rare but GB-reactive $V\delta 2^-$ $\gamma\delta$ TILs. Notably, unintentional activation of $V\delta 2^-$ $\gamma\delta$ T cells has been described in MHC class I-deficient colorectal cancer.¹³ Although we did not observe substantial downregulation of class I molecules in GB, in contrast to reports by others,^{7,42} our findings imply that alternative tumor escape mechanisms may underlie the impaired $\alpha\beta$ TIL reactivity while also raising the possibility that immune checkpoint inhibition could selectively engage at least $V\delta 2^-$ $\gamma\delta$ TILs. However, checkpoint inhibitor trials in GB have largely failed,⁴³ likely due to the overall low prevalence of responsive $V\delta 2^-$ $\gamma\delta$ T cells and lack of appropriate patient selection. Additionally, inhibitory mechanisms specific to $V\delta 2^-$ $\gamma\delta$ T cells in GB remain poorly characterized and may further limit their therapeutic potential.⁴⁴

$V\delta 2^+$ $\gamma\delta$ T cells emerged as the more promising effector population within GB. Although their frequency in GB tissue was lower than that seen in peripheral blood of healthy individuals, they were the dominant $\gamma\delta$ TIL subset and exhibited higher cytotoxic activity when compared with autologous PBMC-derived $V\delta 2^+$ $\gamma\delta$ T cells. A higher proportion of $V\delta 2^+$ $\gamma\delta$ TILs in our cohort was linked to longer overall survival in patients. However, the antitumor activity of $V\delta 2^+$ $\gamma\delta$ TILs varied greatly among individuals, reflecting pronounced inter-patient heterogeneity in $\gamma\delta$ T cell-mediated tumor recognition. This enhanced yet variable activity may be driven by multiple mechanisms, including lower expression of the inhibitory receptor NKG2A in some patients and higher levels of the activating receptor DNAM-1. However, in contrast to earlier reports,⁴⁵ we did not find consistent functional evidence for DNAM-1 involvement in GB recognition, possibly due to incomplete assessment across our full cohort. Instead, reactivity appeared to rely on other activating receptors,

such as NKG2D, NKp46, and NKp44, whose contributions varied markedly between patients. These findings highlight the inter-patient heterogeneity in receptor expression and effector mechanisms that influence the anti-GB activity of $V\delta 2^+$ $\gamma\delta$ TILs, emphasizing that only a subset of patients benefits from higher $V\delta 2^+$ $\gamma\delta$ TIL numbers, underscoring the importance of both quantity and functional quality.

Our study also offers mechanistic insights to enhance $V\delta 2^+$ $\gamma\delta$ TIL activity. First, we observed variable functional impact of inhibitory receptors such as PD-L1 and TIGIT on $V\delta 2^+$ $\gamma\delta$ T cell-mediated responses across patients, indicating that checkpoint blockade would require combinatorial and personalized approaches.⁴³ Second, all responsive $V\delta 2^+$ $\gamma\delta$ TILs depended on $\gamma\delta$ TCR activation, yet tumor recognition was minimal without the addition of aminobisphosphonates, suggesting incomplete assembly of the BTN2A1/BTN3A ligand complex on GB cells.

Although over 90% of GBs show activation of the PI3K/Akt/mTOR pathway,⁴⁶ which promotes BTN2A1 expression,¹⁸ this alone was insufficient to fully assemble the $V\delta 2^+$ $\gamma\delta$ TCR ligand complex. Increased intracellular phosphoantigen levels, enhanced by bisphosphonates, are required to induce BTN3A conformational changes necessary for $\gamma\delta$ TCR activation.^{11,47} We found that BTN3A expression or surface localization was a limiting factor in many patient tumors. Thus, achieving therapeutic concentrations of bisphosphonates at the tumor site is essential. However, systemic administration, standard in clinical trials,^{48–52} may be suboptimal for GB, given that bisphosphonates preferentially accumulate in bone and show poor enrichment in the central nervous system.⁵³ Intrathecal delivery or nanoparticle-based formulations capable of crossing the blood–brain barrier⁵⁴ could provide more effective alternatives. Furthermore, sequential combination with personalized checkpoint inhibitors may synergize with $V\delta 2^+$ $\gamma\delta$ TIL activation to enhance efficacy.

Finally, our transcriptomic analysis identified tumor-intrinsic pathways contributing to $\gamma\delta$ T cell suppression. Among three functional hits, TGF β 1, LAMA5, and ITGA3, TGF β is a well-established immune suppressor.⁵⁵ LAMA5, a component of the extracellular matrix (ECM), has been implicated in cancer cell survival⁵⁶ and was found to suppress $V\delta 2^+$ $\gamma\delta$ TIL activity in our functional assays. Although LAMA5 interacts with ITGA3,⁵⁷ a receptor expressed by GB cells, we found that LAMA5 knockout did not alter BTN2A1/BTN3A expression, suggesting additional suppressive mechanisms. In contrast, ITGA3 knockout increased BTN2A1 and BTN3A, suggesting that ITGA3 may indirectly modulate $V\delta 2^+$ $\gamma\delta$ T cell recognition by controlling ligand availability. Moreover, ITGA3 overexpression correlated with poor survival in GB patients within the TCGA dataset, validating its potential as a therapeutic target.^{58–61} LAMA5 has also been associated with poor prognosis in multiple other cancers,^{62–65} further supporting the relevance of the LAMA5-ITGA3 axis in immune escape.

In conclusion, our study demonstrates that $V\delta 2^+$ $\gamma\delta$ TILs in GB are capable of recognizing and killing autologous tumor cells, and that higher frequencies of these cells are associated with improved overall survival. However, their efficacy is constrained by both immune-intrinsic and tumor-intrinsic suppressive mechanisms, as well as marked inter-patient heterogeneity in receptor usage and effector function. Critically, our use of a fully autologous system revealed a high degree of inter-patient heterogeneity that would be masked in conventional models, reinforcing the need for personalized approaches. Rather than simply increasing $\gamma\delta$ T cell numbers, we propose strategies to enhance their functional potency by ensuring optimal ligand presentation (BTN2A1/BTN3A), overcoming suppressive signals (eg, via LAMA5-ITGA3 blockade), and tailoring immunomodulatory interventions. Stratifying patients by $\gamma\delta$ TIL infiltration and ligand expression, and optimizing intratumoral drug delivery, will be essential to unlocking the therapeutic potential of $\gamma\delta$ T cells in GB.

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Competing interests JK is a cofounder and shareholder of Gadeta Founder BV. JK, DXB and ZS are inventors on patents with $\gamma\delta$ TCR/ligand related topics.

Patient consent for publication Not applicable.

Ethics approval All patients participating in this study have given informed consent to the use of primary tumor and blood samples for research purposes according to the 09-420 Targeted and Individualized Treatment for Malignant Gliomas (TITMAG), and 19-101 Brain Innovative Models and Vectors for ImmunoTherapy (IMOVEIT) protocols, approved by the Ethics Committee of the University Medical Center Utrecht.

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Data availability statement Data are available on reasonable request. Correspondence and requests for raw data or materials should be addressed to JK.

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